TICKETÉ

Product Requirements Document (PRD)

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Item	Detail
Product	Event ticketing platform for organizers, volunteers and attendees
Market	Bangladesh (Asia/Dhaka time zone, BDT currency, mobile banking payments)
Release	Version 1.0 with dummy payment gateway
Tech	Django + Django REST Framework (backend), Next.js (frontend), PostgreSQL on Supabase

1. Overview and Problem

Small and mid-size organizers (campus fests, concerts, cultural days, workshops) usually sell tickets through chat messages, paper slips or spreadsheets. This causes duplicate or fake tickets, long entry queues, no live sales numbers and manual reporting. The platform gives organizers a self-service way to publish an event, sell tickets online, validate each ticket exactly once at the gate and see results afterward.

2. Goals and Non-goals

Goals

- An organizer can go from sign-up to a live, approved event link in under 15 minutes.
- A buyer can complete a purchase in under 2 minutes, with or without an account.
- No ticket can be oversold and no ticket can be admitted twice.
- Volunteers can check in an attendee in under 3 seconds per scan.
- Organizers see live sales and a final report without manual exports.

Non-goals (V1.0)

- Refunds, cancellations workflow, payouts, platform commission (organizer handles refunds directly).
- Real payment gateway (dummy only, but architecture must allow swapping).

- Native mobile apps and reserved seat maps.

3. Users and personas

Persona	Needs	Pain today
Organizer	Create event fast, sell online, know sales live, control who can scan	Manual Google Forms and bKash screenshots; fake tickets
Volunteer	Simple scanner, clear valid/invalid result	Paper lists, slow checks, arguments at the gate
Buyer	Quick purchase, ticket on phone, option to gift a friend	Needs to message organizers, waits for confirmation
Admin (platform)	Review events quickly, stop abuse	No visibility into what is being sold

4. Functional Requirements (FRs)

Priority: M = must (MVP), S = should (V2.0), C = could (later).

4.1 Authentication

ID	Requirement	Pri
FR-A1	Users can register and log in with email and password; email must be verified before an organizer can submit an event.	M
FR-A2	Password reset by emailed link.	M
FR-A3	Roles: Admin, Organizer, Buyer; team roles (Manager, Scanner, Viewer) are assigned per event.	M
FR-A4	Google sign-in.	S

4.2 Event creation

ID	Requirement	Pri
FR-E1	Organizer creates an event with: name, description, venue, banner image, event date and time, sales start and end.	M
FR-E2	Organizer defines one or more ticket categories, each with name, price (BDT) and seat limit (blank = unlimited).	M
FR-E3	Organizer sets maximum tickets per buyer per order.	M
FR-E4	Organizer can enable 'Attendee name per ticket'. When on, checkout shows Attendee 1 name, Attendee 2 name... matching the quantity.	M
FR-E5	Organizer can add custom fields (text, number, dropdown, checkbox) and mark each required or optional. Definitions are stored as flexible structured data.	M
FR-E6	Event with no custom fields only asks for quantity, email, payment.	M

FR-E7	Draft saving and public-page preview before submission.	M
FR-E8	After approval, edits to date, venue or prices are restricted; allowed changes notify existing ticket holders.	S

4.3 Admin approval

ID	Requirement	Pri
FR-D1	Submitting an event creates an in-app and email notification for admins.	M
FR-D2	Admin can approve or reject. Rejection requires a reason, emailed to the organizer; organizer can edit and resubmit.	M
FR-D3	On approval, a unique public URL (slug) is generated.	M
FR-D4	Admin can suspend an event or organizer; all admin actions are logged.	S

4.4 Purchase flow

ID	Requirement	Pri
FR-P1	Public event page is viewable without login, server-rendered with title, banner and description for link previews.	M
FR-P2	At checkout the buyer chooses 'Continue as guest' or 'Login'. If logged in, the email field is prefilled.	M
FR-P3	Buyer selects category and quantity, limited by remaining seats and the per-buyer limit.	M
FR-P4	On proceeding to payment, seats are held for 10 minutes; expired holds release automatically.	M
FR-P5	Gifting: buyer may enter a different recipient email. The recipient receives the ticket PDF; the buyer sees a confirmation screen and receives a receipt email.	M
FR-P6	Sold-out categories show 'Sold out' and cannot be selected.	M
FR-P7	Promo codes and waitlist.	C

4.5 Payment (dummy)

ID	Requirement	Pri
FR-Y1	Buyer selects a mobile banking option (bKash, Nagad, etc.), enters a phone number, then an OTP.	M
FR-Y2	Successful OTP confirms the payment and the order; failure or timeout cancels the order and releases the hold.	M
FR-Y3	Payment is accessed through a gateway interface so a real provider can replace the dummy without changing order or ticket logic.	M
FR-Y4	Every payment attempt is stored with status, amount and reference. Funds are treated as received by the organizer.	M

4.6 Tickets, PDF, email

ID	Requirement	Pri
FR-T1	Each purchased ticket gets a unique human-friendly ticket ID and a unique secure token; both are stored with order, event, category, attendee name, recipient email, status.	M
FR-T2	A QR code encoding the secure token (not a sequential ID) is generated for each ticket and stored.	M
FR-T3	One PDF per order, one page per ticket, using the fixed design in section 7.	M
FR-T4	PDF is emailed from the platform address to the recipient and is downloadable on the confirmation page. Email and PDF generation run in the background.	M
FR-T5	Guests can retrieve tickets through 'Find my ticket' using email plus an emailed OTP and can request a resend.	M
FR-T6	Registered buyers see an order history page.	S

4.7 Check-in

ID	Requirement	Pri
FR-C1	Scanner page (mobile web) reads QR via camera; manual ticket ID entry is available as a fallback.	M
FR-C2	On first valid scan, the ticket becomes Used with timestamp and scanner identity. The operation is atomic, so simultaneous scans admit only one.	M
FR-C3	Result screen: green for valid; red with reason for already used (time, scanner), invalid, wrong event or cancelled. Shows attendee name and category.	M
FR-C4	Live counter of checked-in vs total tickets.	M
FR-C5	Downloadable attendee list for offline fallback.	S

4.8 Team management

ID	Requirement	Pri
FR-M1	Owner invites teammates by email; the invitation contains a link that expires (7 days) and must be accepted.	M
FR-M2	Roles per event: Manager (edit, dashboard, reports, attendee list), Scanner (check-in only), Viewer (read-only reports).	M
FR-M3	Owner can change a role or revoke access at any time.	M
FR-M4	Activity trail of check-ins by team member.	S

4.9 Dashboard and reports

ID	Requirement	Pri
FR-R1	Basic dashboard charts: tickets sold over time, sales by category, revenue, check-in progress.	M
FR-R2	Final report after closing: total tickets sold and unsold, per category, revenue, checked-in vs no-show.	M
FR-R3	CSV export of orders and attendees; report as PDF.	S

5. Key Business Rules

Rule	Description
Seat limits	Limits are per category. Event capacity is the sum of categories. Blank limit means unlimited.
No overselling	Availability check and hold creation happen in one database transaction with row locking.
Hold timeout	10 minutes; expired holds return seats to the pool.
Single use	A ticket can be marked Used only once; the check is atomic.
Ticket ownership	Ticket belongs to the recipient email. In a gift, buyer and recipient differ.
Per-buyer limit	Applies per order and per email for the same event.
Time	Stored in UTC, shown in Asia/Dhaka. Sales start and end are separate from event time.
Refunds	Outside the platform. Organizer may mark a ticket Cancelled so it fails at the gate.
Visibility	Only an event's team and admins can see its buyers and attendees.

6. User Stories with Acceptance Criteria

Story	Acceptance criteria
As an organizer, I create an event and ticket categories so I can sell tickets.	Form validates required fields; at least one category with price; saved as Draft; submit moves to Pending approval.
As an admin, I approve or reject events so only legitimate events go live.	Pending list visible; reject needs reason; approve generates working public link; organizer is emailed.
As a buyer, I purchase as guest so I do not need an account.	Checkout works with email only; PDF arrives in that email; confirmation page offers download.
As a logged-in buyer, my email is prefilled.	Email field populated and editable only when gifting is selected.

As a buyer, I buy 3 tickets with attendee names.	Three name boxes appear when enabled; one PDF with 3 pages; each page has own QR and name.
As a buyer, I gift a ticket to a friend.	Recipient email receives PDF; buyer sees confirmation and receipt.
As a volunteer, I scan tickets quickly.	Valid QR shows green and marks Used; repeat shows red with time and scanner.
As an organizer, I invite volunteers.	Invitee gets email link; after accepting, sees only role-permitted pages.
As an organizer, I view sales after the event.	Report shows sold, unsold, per category, revenue, attendance.

7. Ticket PDF Specification

The ticket design is fixed by the platform.

Format: A4 portrait; one page per ticket.

Zone	Content
Header	Platform logo and name; event name in large type.
Banner	Event banner image (cropped to fit).
Details block	Date and time (Asia/Dhaka), venue, ticket category, attendee name (if collected).
Identity block	Ticket ID in readable text, order number, ticket number 'X of Y'.
QR zone	Large QR code (minimum 35 mm) with 'Scan at entry' label.
Footer	Terms line (one use only, keep QR private), organizer contact, platform support email.

Design to be finalized with a visual mock-up before development of the PDF module.

8. Non-functional Requirements (NFRs)

Area	Requirement
Performance	Event page loads under 2 seconds on a typical mobile connection; scan result under 1 second after request; ticket email within 1 minute of payment.
Scalability	Handle a sales spike (for example, 500 buyers within a few minutes) without overselling.
Security	Passwords hashed; tokens in httpOnly cookies; secure random QR tokens; rate limiting on login, OTP, purchase; upload validation; role checks on every endpoint.

Reliability	Emails and PDFs retry on failure; payment and order states are consistent after failures.
Privacy	Buyer data accessible only to the relevant event team; unnecessary data not stored.
Usability	Mobile-first; scanner and checkout usable on small phones; clear error messages.
Accessibility	Readable contrast, labeled form fields, keyboard navigable forms.
Observability	Error logging and an audit log for admin actions and check-ins.

9. Technical Approach

Layer	Choice and notes
Frontend	Next.js on Vercel. Server-rendered public event page for share previews; client-side scanner page with camera QR library; form library for dynamic organizer forms; charting library for dashboard.
Backend	Django + Django REST Framework. JWT authentication (httpOnly cookies). Django handles users; Supabase is used only for database and storage.
Database	PostgreSQL on Supabase. Transactions and row locking for seat holds and check-in; JSON column for custom form fields. Check pooler connection mode.
Background jobs	Celery with Redis (or lighter alternative) for emails, PDF generation and hold expiry.
PDF and QR	WeasyPrint (HTML/CSS ticket design) and a Python QR library.
Email	Transactional provider (Brevo, Resend or SendGrid) sending from the platform address.
File storage	Supabase Storage for banners and generated PDFs.
Hosting	Frontend on Vercel; Django and workers on Render, Railway, Fly.io or a VPS, in a region near Bangladesh users.
Repository	One monorepo with /backend and /frontend folders (can be split later).

Main Data Entities

Entity	Key fields/relations
User	email, password hash, name, role (admin/organizer/buyer)
Event	organizer, name, description, venue, banner, start/end time, sales start/end, status, slug, max per buyer, attendee-name flag
EventFormField	event, label, type, required, options (flexible structure)
TicketCategory	event, name, price, seat limit, sold count
TeamMember/Invitation	event, email, role, status, token, expiry

SeatHold	event, category, quantity, expires at, order link
Order	event, buyer (or guest email), recipient email, total, status, custom field answers
Payment	order, provider, phone, amount, status, reference, OTP result
Ticket	order, category, ticket ID, secure token, attendee name, status, used at, used by
Notification/AuditLog	recipient, type, message, read state; actor, action, target, time

10. Success Metrics

Metric	Target for pilot events
Checkout completion rate	60% or higher of buyers who start payment
Oversold tickets	0
Duplicate admissions	0
Ticket email delivered within 1 minute	95% or higher
Average check-in time	3 seconds or less
Admin approval turnaround	Within 24 hours

11. Risks and Mitigations

Risk	Mitigation
Overselling in a rush	Transactional locking, load test before pilot.
Poor internet at the venue	Offline attendee list (V2.0), lightweight scanner page.
Emails landing in spam	Transactional provider, verified sender domain.
Fake or shared tickets	Secure tokens, single-use check, ticket status visible to scanners.
Free-tier limits (Supabase, hosting)	Use paid plan for production events.
Replacing dummy gateway later	Gateway interface from day one.

12. Milestones

Phase	Scope	Rough duration
1	Auth, event form, categories, approval, public page	1 week
2	Holds, order, dummy payment, ticket, QR, PDF, email	1 week
3	Scanner, single-use check, team invites and roles	1 week

4	Dashboard, final report, find-my-ticket, testing and polish	1 week

13. Open Questions

ID	Question	Owner
OQ-1	Final ticket PDF layout and branding (logo, colors)?	Product / Design
OQ-2	Should the buyer also receive the PDF when gifting or only a receipt?	Product
OQ-3	Are free (zero price) tickets allowed?	Product
OQ-4	Minimum notice before event date when submitting for approval?	Admin
OQ-5	Languages: English only or Bangla as well?	Product
OQ-6	Does the organizer see buyers' phone numbers or only emails?	Product / Legal